Question Bank

Module-4

Q1 A college conducts both morning and evening classes intended to be identical. A sample of 100 morning students yields examination results as under:

Mean = 72.4, S.D = 14.8.

A sample of 200 evening students yields examination results as under:

Mean = 73.9, S.D = 17.9.

Is there any significant difference between the mean results of both the classes? Test at 5% level of significance.

[Ans: Not significant]

Q2 A manufacturer claims at least 95% of the items he produces are failure free. Examination of a random sample of 600 items showed 39 to be defective. Test the claim at a significance level of 0.05.

[Ans: Significant]

Q3 In a sample of 500 people in Kerala, 280 are tea drinkers and the rest are coffee drinkers. Can we assume that both coffee and tea are equally popular in this state at 1% level of significance?

[Ans: Not Significant]

Q4 The mean weekly sales of the chocolate bar in a candy stores was 140.3 bars per store. After an advertisement campaign, the mean weekly sales in 22 store for a typical week increased to 153.7 and showed a standard deviation of 17.2. Was the advertising campaign not successful?

Ans. $t = 1.9716$; the advertisement was successful at 5% level of significance.

Q5 Talcum powder is packed into tins by a machine. A random sample of 11 tins is drawn and their content are found to weigh in kgs as follows: 0.44, 0.51, 0.49, 0.52, 0.45, 0.48, 0.46, 0.45, 0.47, 0.45 and 0.47. test if the average packing can be taken to be 0.5 kgs.

[Ans. $t = 3.43; Not\ Significant$]

Q6 The incomes of a random sample of laborers in the industry I are: Rs. 630, 650, 680, 690, 710, and 720 per month. The incomes of a similar sample from Industry II are Rs. 610, 620, 650, 680, 690, 700, 710, 720, are 730 per month. Discuss the validity of the suggestion that the industry I pays labourers much better than the industry II.

[Ans. t= 0. 196.There is no significant difference between the salaries of Industry **I** and Industry **II.]**

Q7 The sales data of an item in six shops before and after a special promotional campaign are as under

Shops	*A*	*B*	*C*	*D*	*E*	*F*
Before campaign	*53*	*28*	*31*	*48*	*50*	*42*
After campaign	*58*	*29*	*30*	*55*	*56*	*45*

Can the campaign be judged to be a success? Test at 5% level of significance.

Ans. $t = -2.78$; the sales campaign has been success.

Q8 Two random samples drawn from normal populations are:

SampleI	20	16	26	27	23	22	18	24	25	19	20
SampleII	17	23	32	25	22	24	28	18	31	33	27

Obtain estimates of the variances of the populations, and test whether the two populations have the same variance. Use F – Test.

	(**Ans**: F=2.14)
Q9	The time taken by workers in performing a job my method I and method II is given below: Method I: 20 16 26 27 22 Method II: 27 33 42 35 32 34 38. **[Ans.** $F = 1.37$, variance of time distribution in a population from which the samples are drawn do not differ significantly.]
Q10	A sample of 400 people with a particular disease was selected, of which 50% were given a drug, 35% were not cured while 56.25% of not cured were not given any drug. Test at 5% level of significance whether the drug is effective or not? [**Ans**. Computed value of $\chi^2 = 2.084$.]
Q11	The number of parts of a particular spare part in a factory was found to vary from day to day. In a sample study the following information was obtained. *Day* | *Mon.* | *Tue.* | *Wed.* | *Thur.* | *Fri.* | *Sat.* | *Sun.* *No. of part demanded* | 1124 | 1125 | 1110 | 1120 | 1126 | 1115 | 6720 Test the hypothesis that the number of parts demanded does not depend on the day of the week. (The table value of x^2 for 5 d.f. at 5% level of significance is 11.07. [**Ans**. $\chi^2 = 0.179.$]
Q12	To test the efficiency of a new drug a controlled experiment was conducted wherein 300 patients were administered the new drug and 200 other patients were not given the drug. The patients were monitored and the results were obtained as follows: *Cured* | *Condition worsened* | *No effect* | *Total* *Given the drug* | *200* | *40* | *60* | *300* *Not given the drug* | *120* | *30* | *50* | *200* *Total* | *320* | *70* | *110* | *500* Use $\chi^2(Chi - Square)$ test for finding the effect of the drug. [**Ans**. Compute value of $x^2 = 2.435$]
Q13	Five coin are tossed 3,200 times and the number of heads appearing each time are noted. At the end, the following results were obtained: *No. of heads* | 0 | 1 | 2 | 3 | 4 | 5 *Frequency* | 80 | 570 | 1100 | 900 | 500 | 50 Use Chi-square test of goodness of fit to determine whether the coins are unbiased. [**Ans**. Computed value of $\chi^2 = 58.8$]
Q14	The theory predicts the production of beans in the four groups A, B, C and D should be 9 : 3 : 3 : 1. In an experiment among 1,600 beans, the numbers in the four groups were 882, 313, 287, and 118. Does the experimental results support the theory? [**Ans**. $\chi^2 = 4.72$]
Q15	*A machine puts out 16 imperfect articles in a sample of 500. After the machine is overhauled, it puts 3 imperfect articles in a batch of 100. Has the machine improved? Test at 5% level of significance*. **[Ans: Not Significant]**